Campus Chakra

Prompt Workbook · Prompt iterations, tools, evidence and human decisions

Purpose. This workbook documents how Campus Chakra was shaped through iterative prompting, implementation and validation. It records only tools and outcomes used in the project; public share links must be added by the team before form submission.

1. Tools and roles

Tool / system	How it supported the work
Codex CLI	Repository inspection, backend/API hardening, smoke tests, documentation and submission assembly.
LLM providers in the product	Optional facilities-copilot explanations with deterministic local fallback when a provider or network is unavailable.
Python / FastAPI / scikit-learn	Data ingestion, validation, edge classification, demand forecasting, dashboard APIs and CSV predictions.
HTML, CSS, JavaScript, Chart.js	Interactive dashboard, facility alerts, allocation views and evaluator-facing guide screens.

2. Prompt iteration log

Iteration	Representative prompt	Decision / outcome
Schema & constraints	“Inspect all P2P dataset tables and map their fields to the problem constraints. Identify late data, aliases and unit conflicts.”	Defined validation, normalization, deduplication and traceable ingestion rules.
Waste classification	“Compare sensor-only and edge-available feature sets; retain a model that meets the accuracy and latency constraints.”	Selected compact forest deployment with explicit sensor-quality features and offline inference.
Forecasting	“Benchmark 6h and 12h demand forecasts on chronological holdout; document accuracy / footprint trade-offs.”	Selected lightweight forecasting bundle and exposes forecast outputs in the API.
Operational copilot	“Ground every recommendation in supplied telemetry, event context and operator notes; supply a deterministic fallback.”	Implemented evidence-led recommendations without making external LLM availability critical.

3. Human contribution and safeguards

- Defined the edge-first architecture, measurable success criteria and evaluator workflow.
- Specified idempotent reconciliation for delayed gateway transmissions.
- Chose fairness-aware, building-normalized incentive logic rather than raw consumption comparisons.
- Added input schema validation so jury uploads fail clearly rather than silently producing invalid results.

4. Public chat share links

Team action required before upload: add genuine public share links below. No link has been invented in this workbook.

Conversation / evidence	Public share link
Prompt-to-product ideation and dataset analysis	TO BE ADDED BY TEAM
Backend validation and feature iteration	TO BE ADDED BY TEAM
UI / demo-video direction	TO BE ADDED BY TEAM